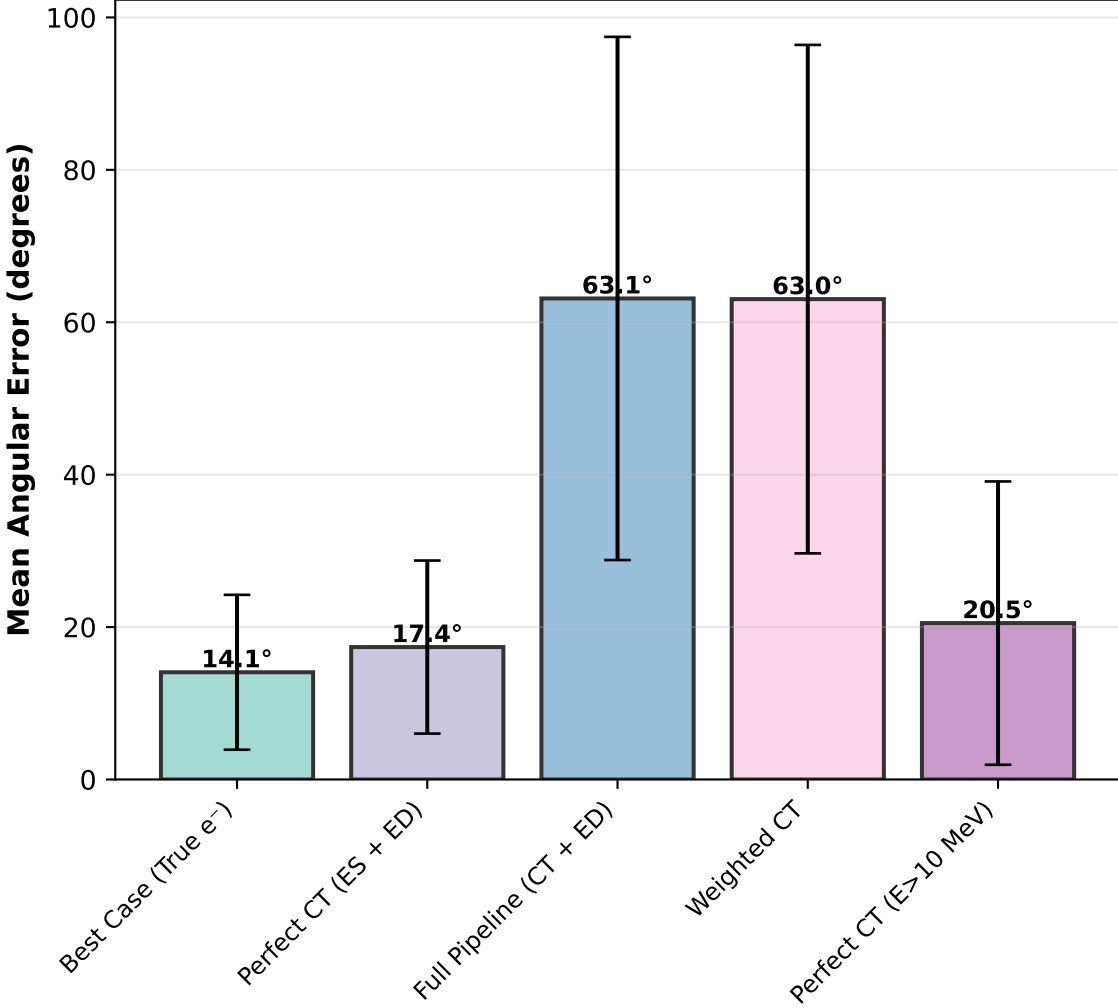


Emcee 6-Scenario Aggregate Analysis

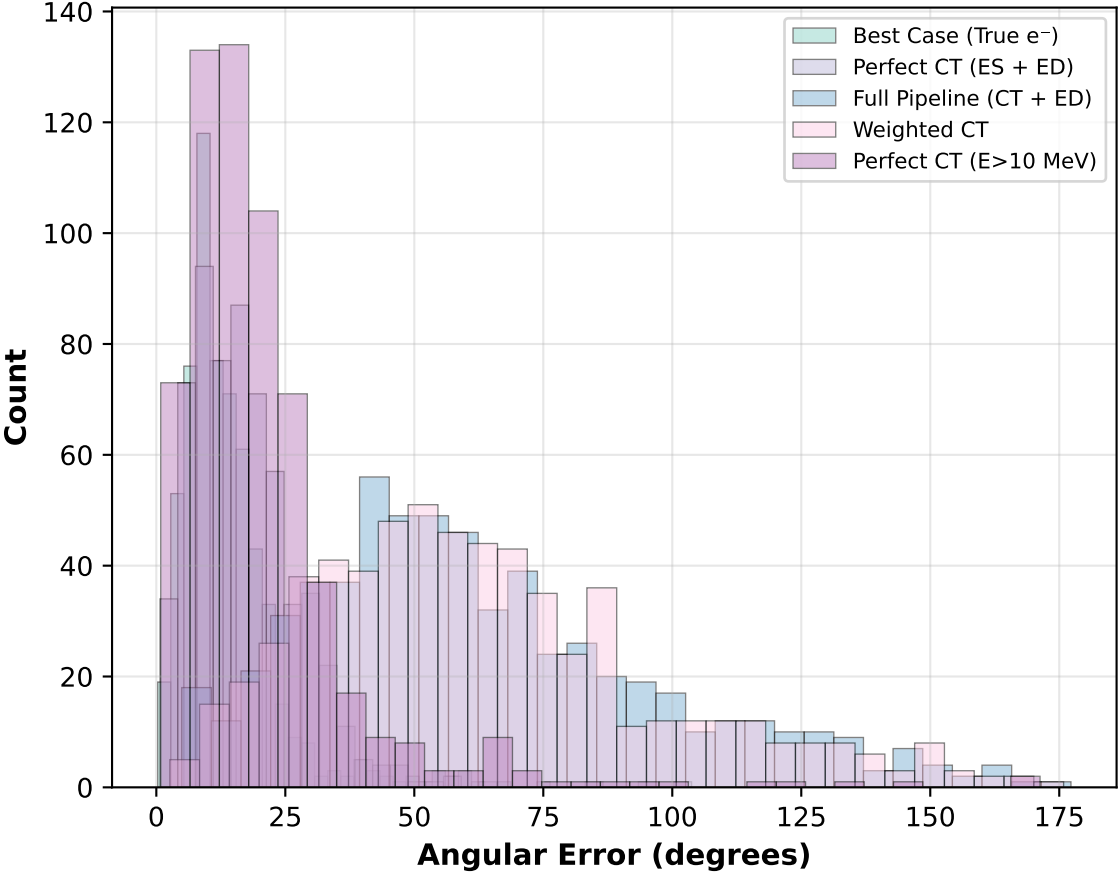
Summary Statistics

Scenario	N	Mean	Median	Std
Best Case (True e ⁻)	615	14.07°	11.67°	10.16°
Perfect CT (ES + ED)	615	17.38°	15.79°	11.35°
Full Pipeline (CT + ED)	618	63.12°	56.61°	34.33°
Weighted CT	618	63.03°	57.63°	33.36°
Perfect CT (E>10 MeV)	615	20.52°	16.40°	18.59°

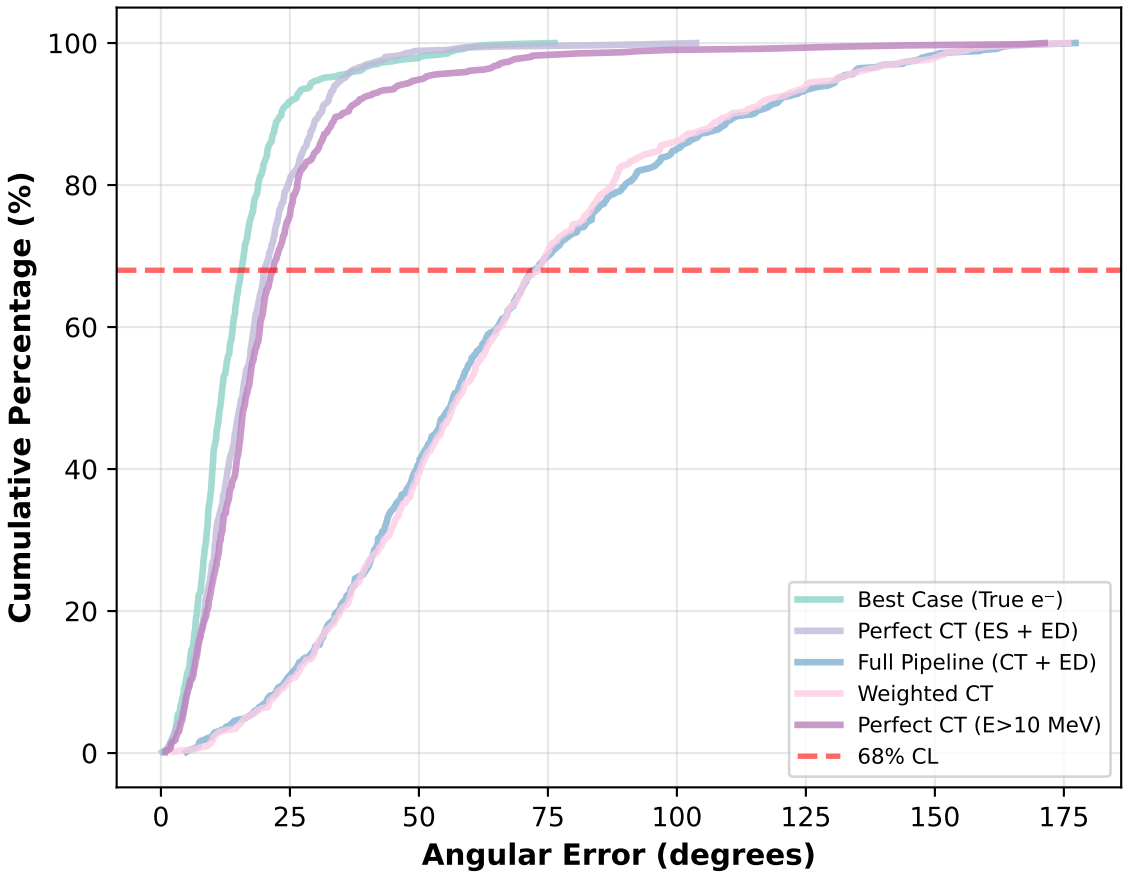
Mean Resolution ± Std Dev



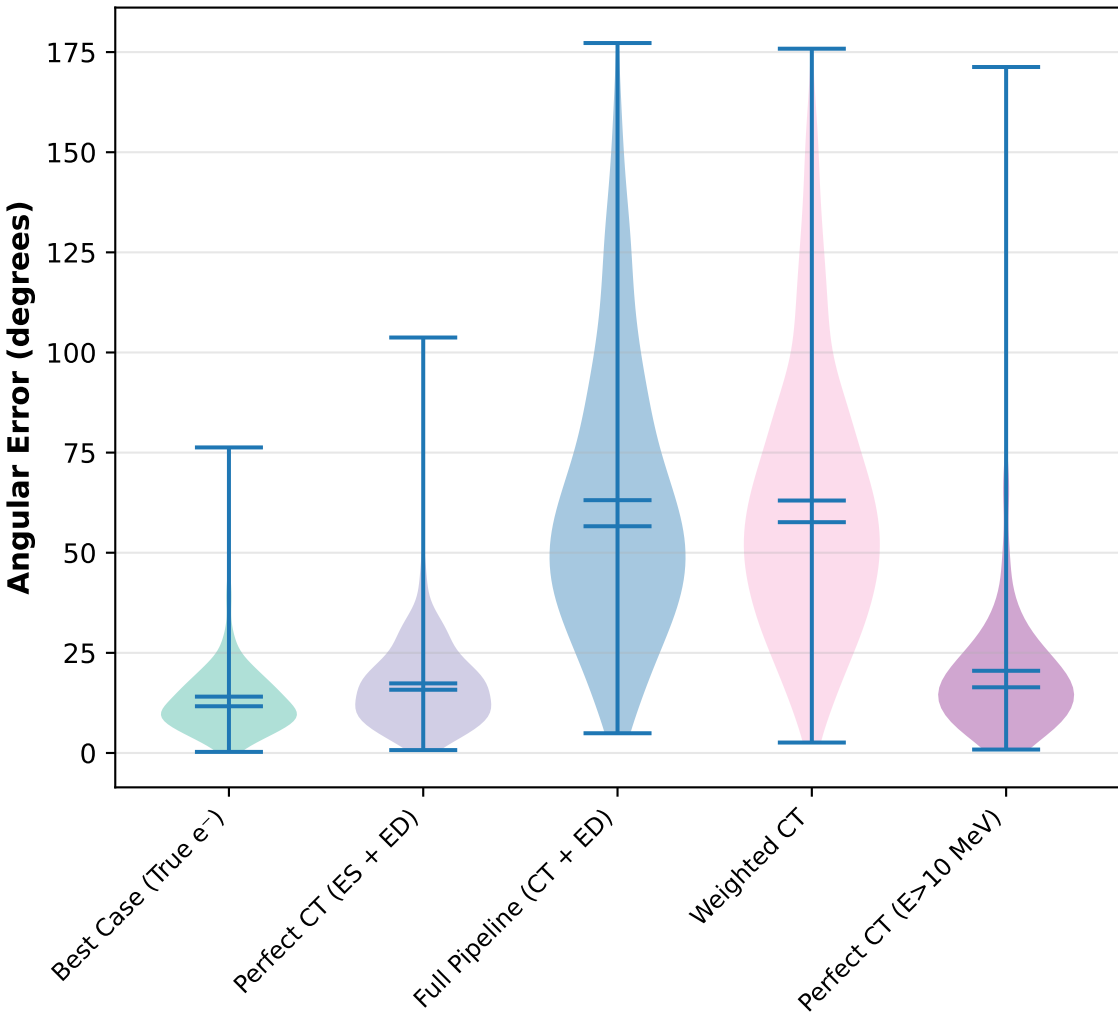
Distribution Overlay



Cumulative Distribution

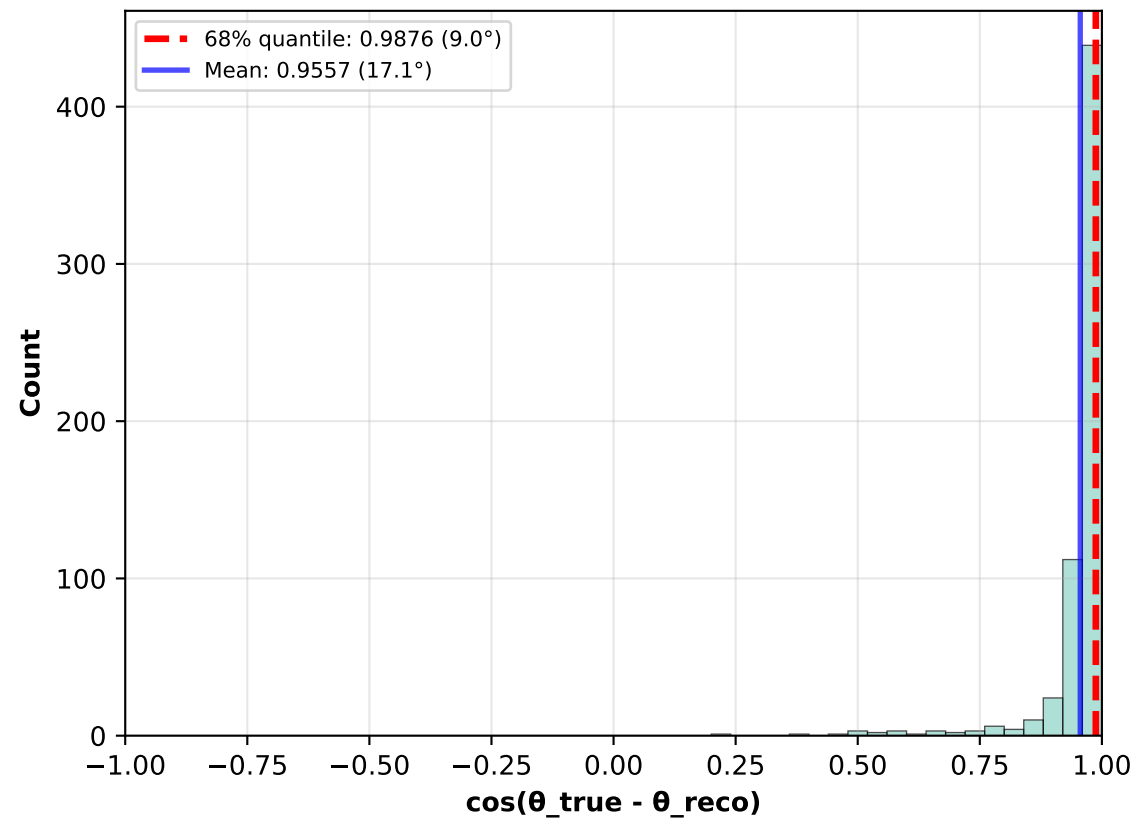


Distribution Shapes

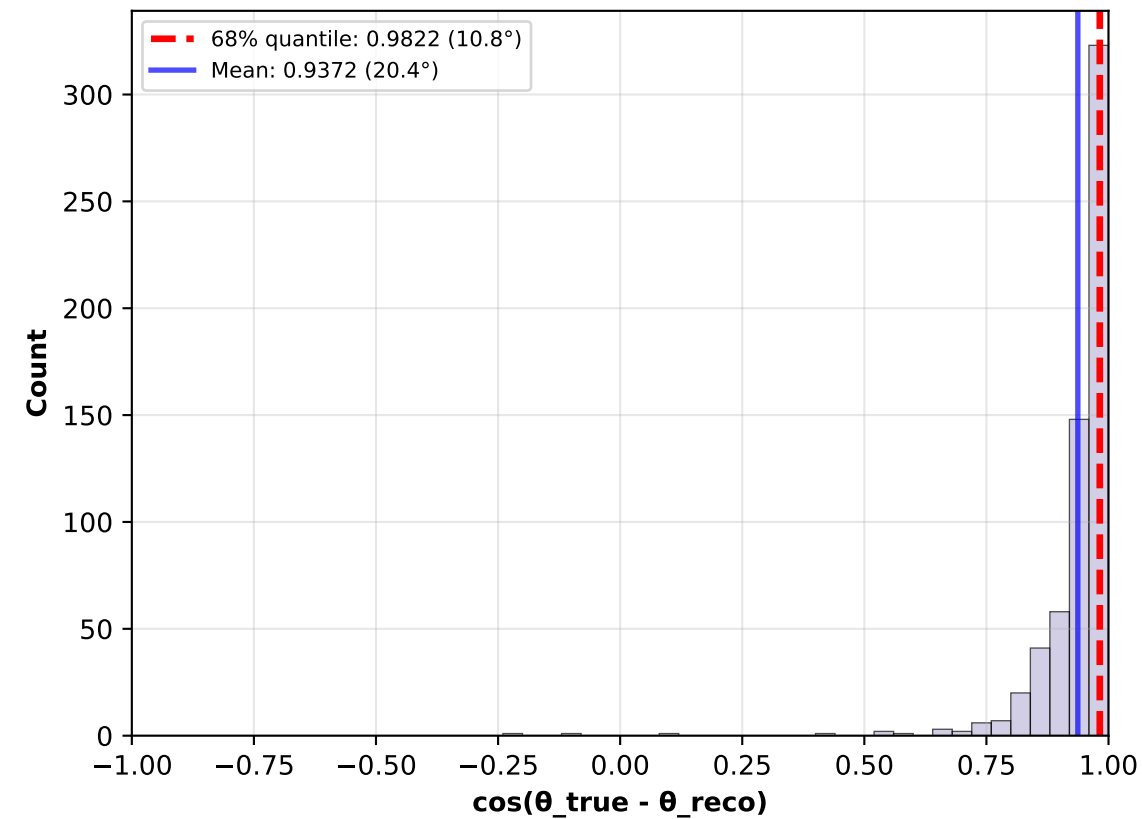


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Full Range

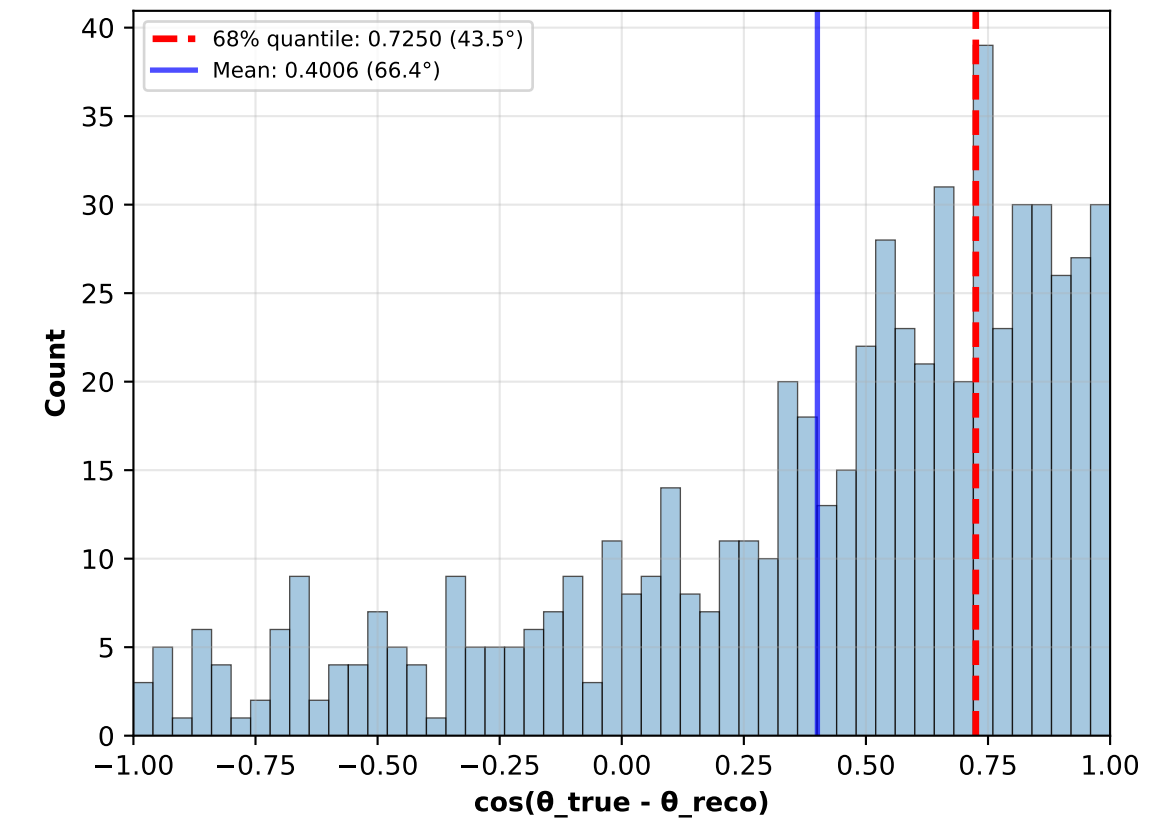
Best Case (True e^-)
(N=615)



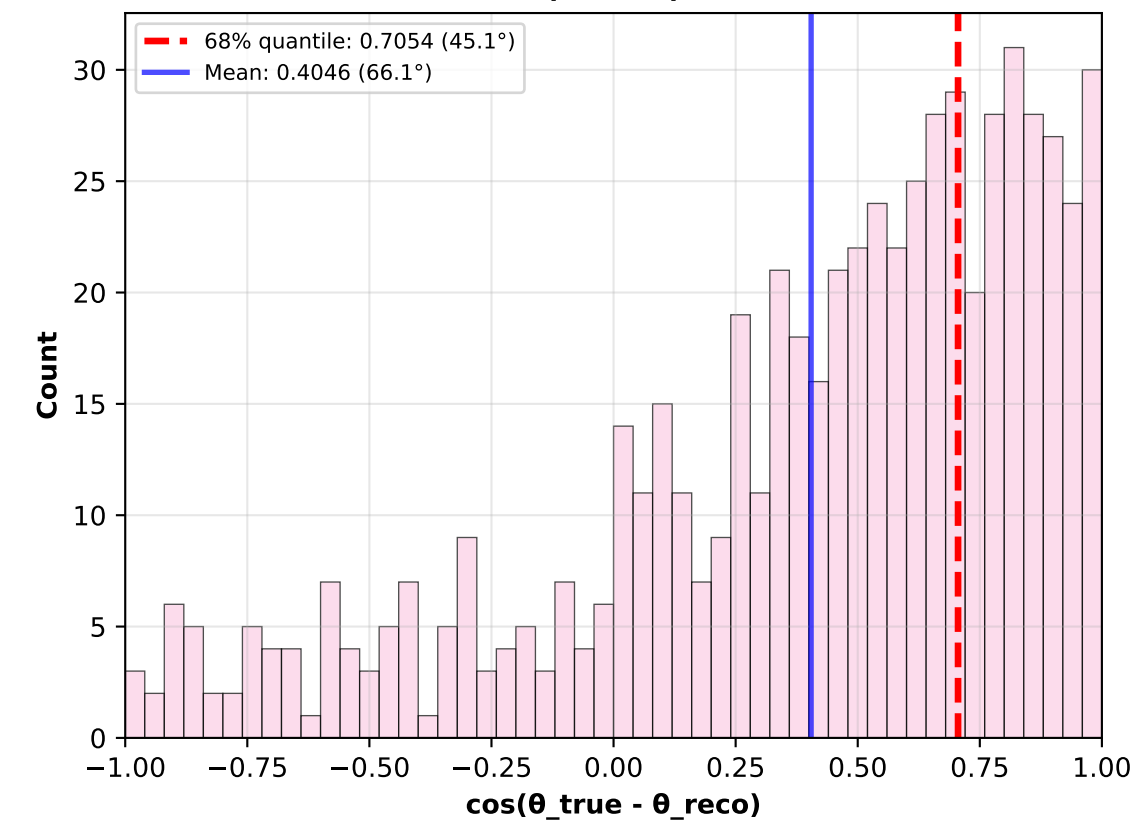
Perfect CT (ES + ED)
(N=615)



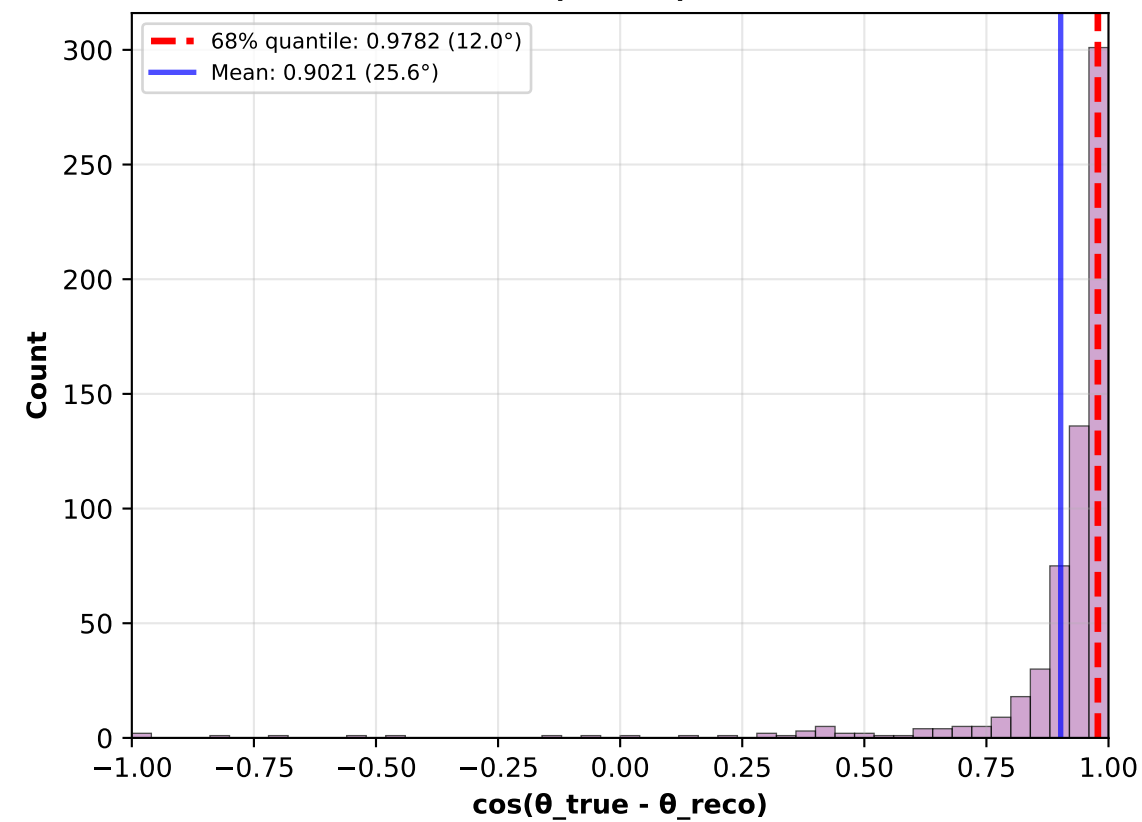
Full Pipeline (CT + ED)
(N=618)



Weighted CT
(N=618)

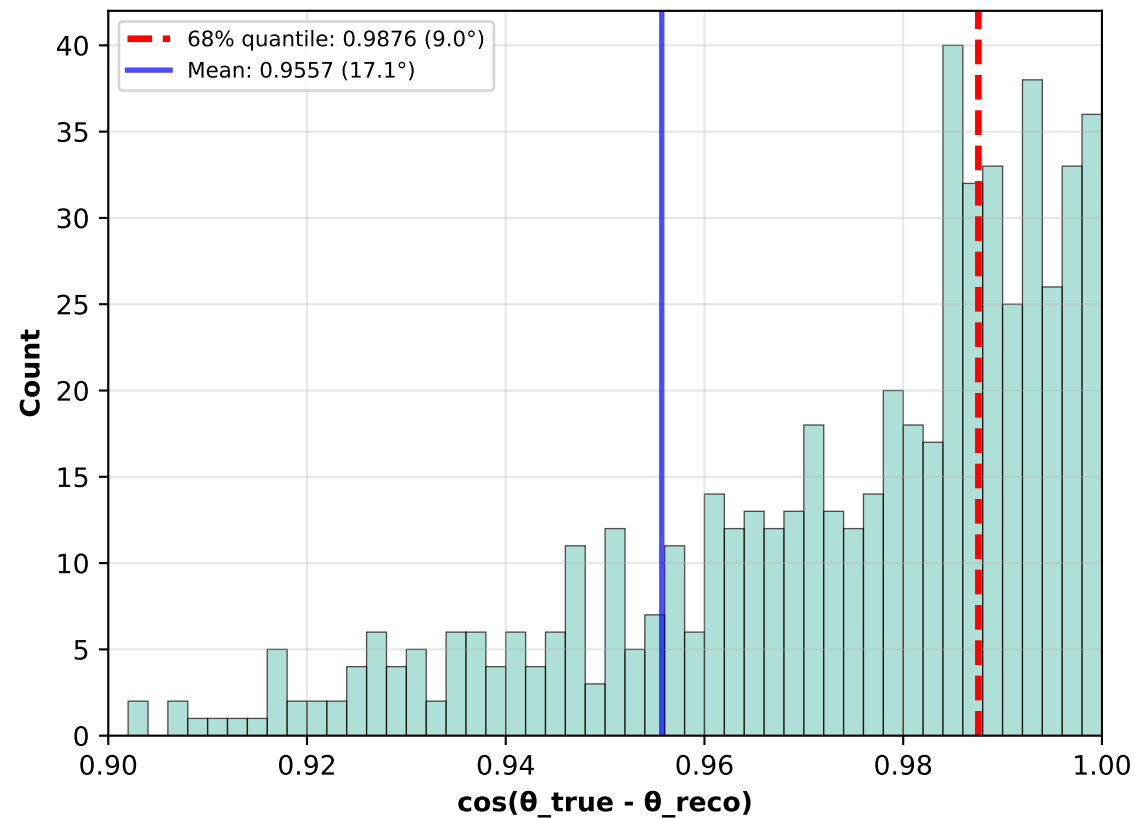


Perfect CT (E>10 MeV)
(N=615)

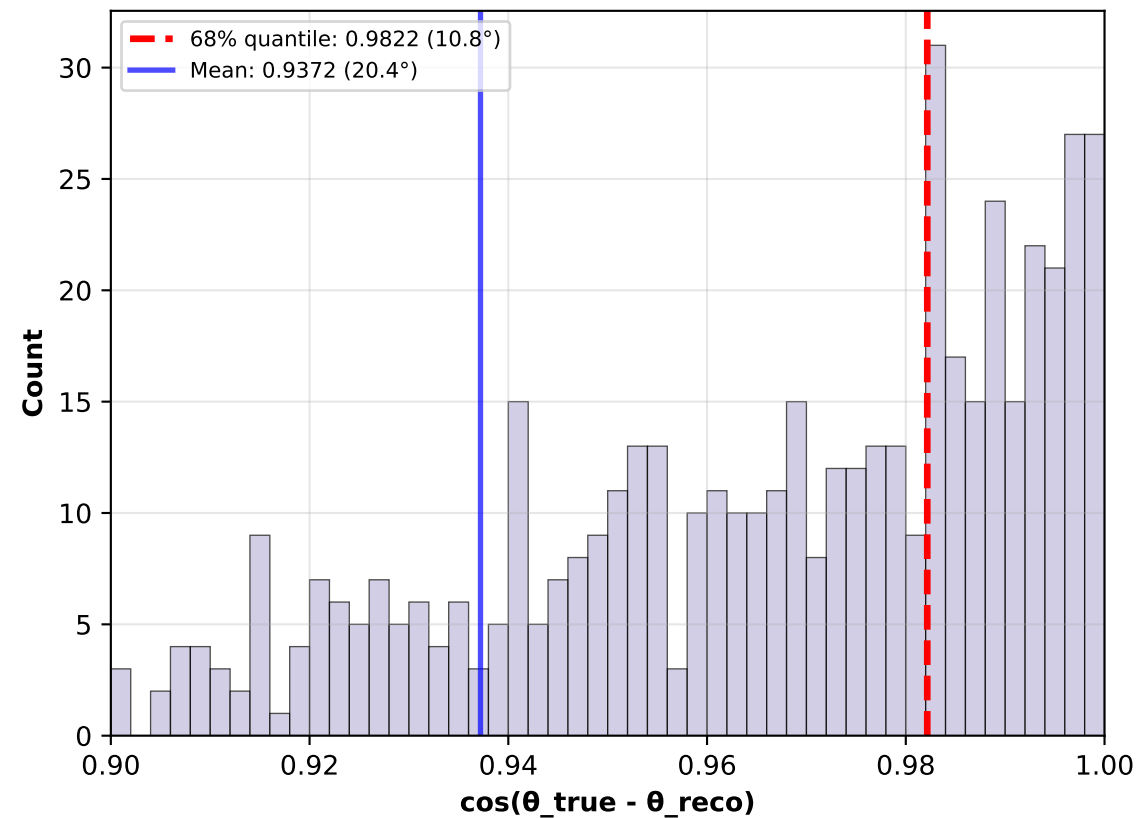


Cosine Distribution: $\cos(\theta_{\text{true}} - \theta_{\text{reco}})$ - Zoomed (0.9-1.0)

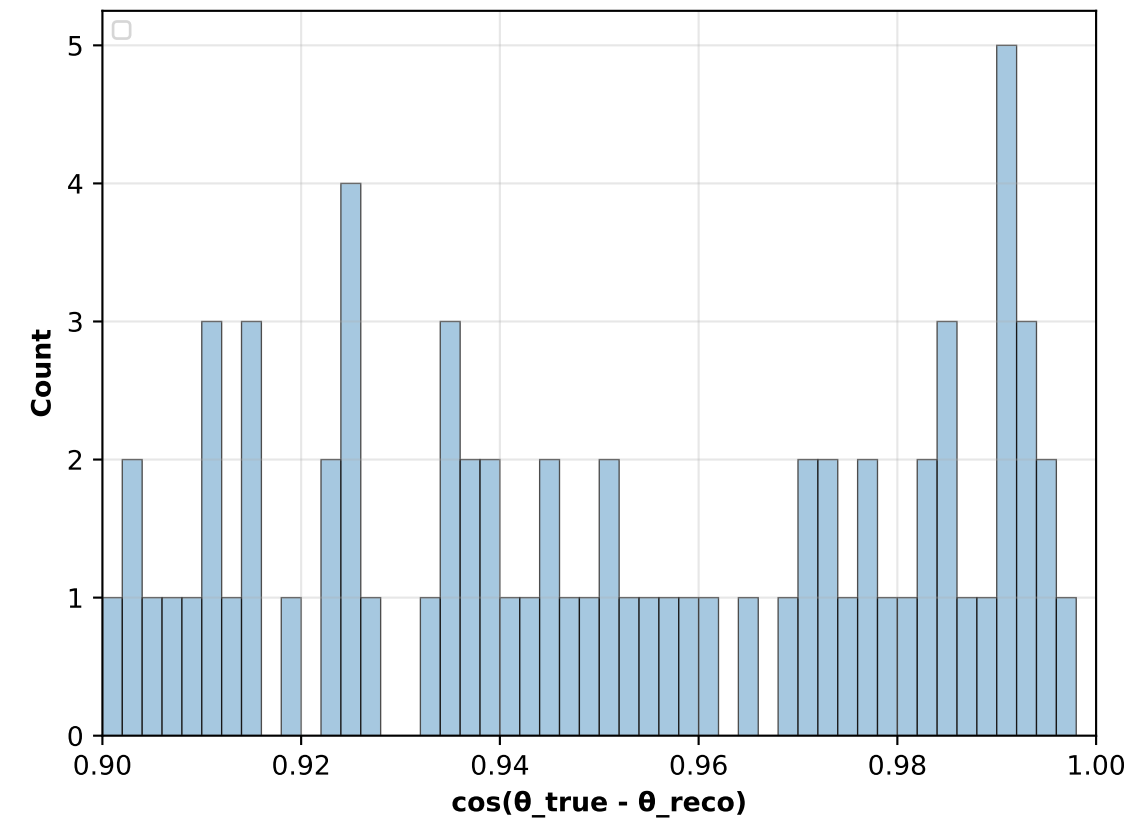
Best Case (True e^-)
(N=615)



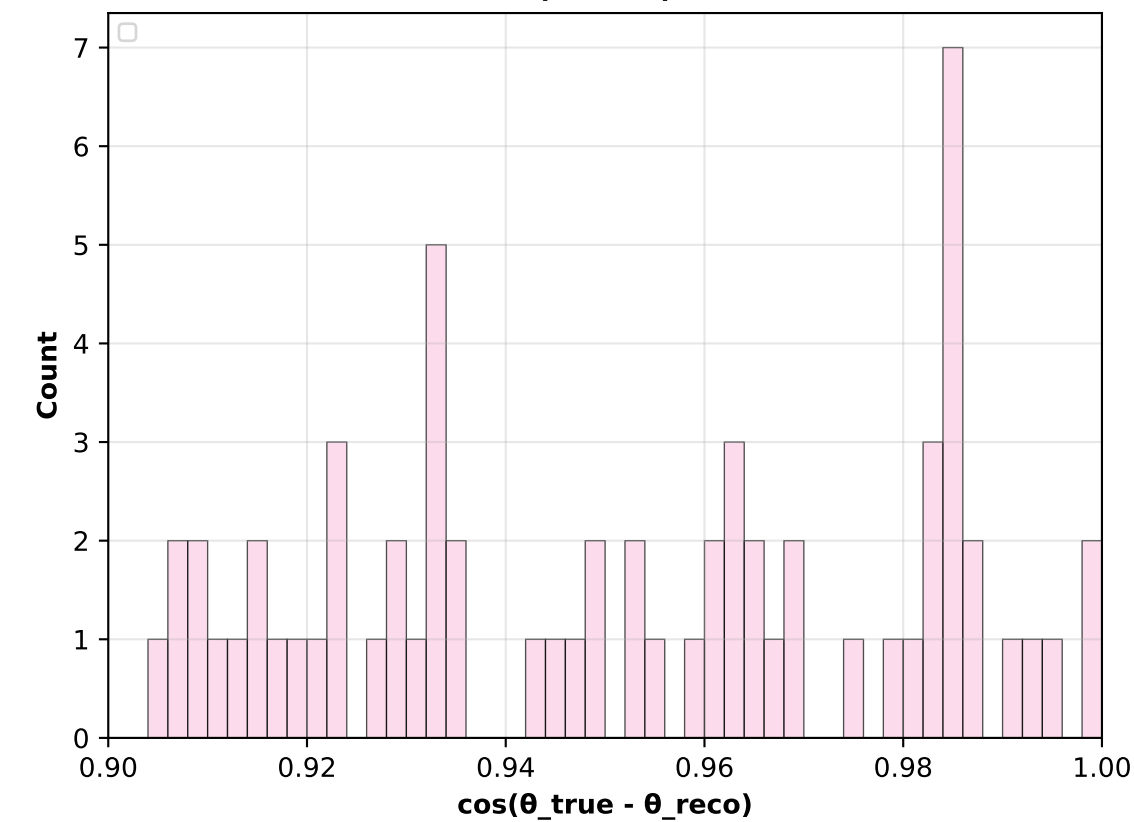
Perfect CT (ES + ED)
(N=615)



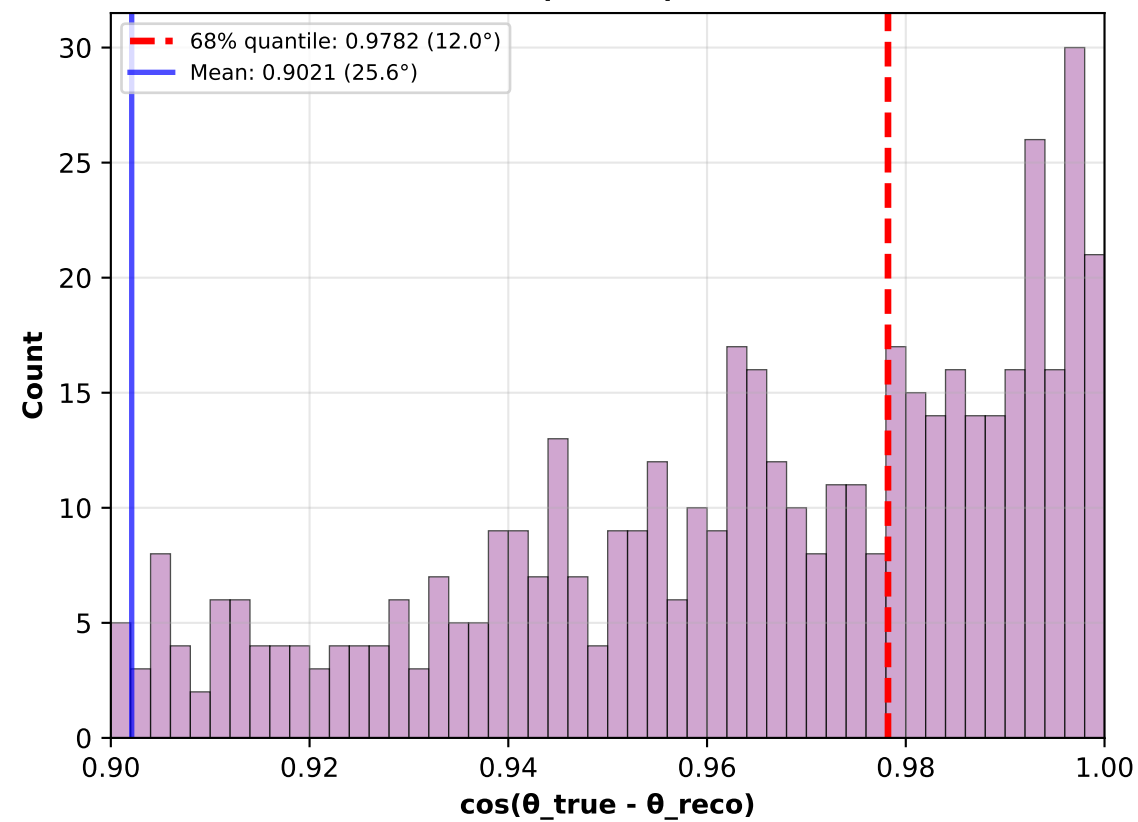
Full Pipeline (CT + ED)
(N=618)



Weighted CT
(N=618)



Perfect CT (E>10 MeV)
(N=615)



Cluster Usage Across Scenarios

Scenario	N Cats	Input Total	Input ES Main	Value "Clusters Used"	Notes
Best Case (True e ⁻)	615	102	10	265	MCMC samples (weighted sum)
Perfect CT (ES + ED)	615	102	10	265	MCMC samples (weighted sum)
Full Pipeline (CT + ED)	618	102	10	3219	MCMC samples (weighted sum)
Weighted CT	618	102	10	3219	MCMC samples (weighted sum)
Perfect CT (E>10 MeV)	615	102	10	144	141.2% retention

Note: "Clusters Used" shows different meanings per scenario:

- Best Case, Perfect CT: Number of ES main clusters actually used (after E>3 MeV cut)
- Full Pipeline, Weighted CT: Weighted sum of cluster contributions in MCMC (includes CT probabilities)
- Energy cut scenarios: Number of clusters passing energy threshold

The key insight: Best Case and Perfect CT use ~85-90% of ES main tracks with E>3 MeV.
Full Pipeline shows much higher values due to weighted MCMC sampling, not actual cluster count.

Input Data Statistics: Files and Clusters per Cat

